EXP. NO: 23

PERFORMANCE EVALUATION OF 8-QAM AND 16-QAM COMMUNICATION SYSTEMS UNDER AWGN USING BER, CONSTELLATION AND EYE DIAGRAM ANALYSIS IN MATLAB

Objectives

1. To generate random binary data and implement 8-QAM and 16-QAM digital modulation in MATLAB.
2. To transmit the modulated signals through an Additive White Gaussian Noise (AWGN) channel for selected SNR/Eb/N0 conditions.
3. To evaluate the bit error rate (BER) performance of 8-QAM and 16-QAM and compare the measured results with the expected trend.
4. To plot and interpret the constellation diagrams of 8-QAM and 16-QAM under different noise conditions.
5. To generate and analyze eye diagrams to observe the effect of AWGN on signal quality and symbol/decision margins.
6. To compare the performance of 8-QAM and 16-QAM in terms of spectral efficiency, BER, constellation separation and eye opening.

CODE

```
clc;
clear;
close all;

% Parameters
M = 16;
N = 1000;
SNR = 10;

% Generate random data
data = randi([0 M-1], N, 1);

% 16-QAM modulation (manual)
levels = [-3 -1 1 3];
I = levels(mod(data,4) + 1);
Q = levels(floor(data/4) + 1);
tx = (I + 1i*Q) / sqrt(10);

% Add AWGN noise
noisePower = 10^(-SNR/10);
noise = sqrt(noisePower/2) * ...
    (randn(N,1) + 1i*randn(N,1));
```

```
rx = tx + noise;

% Demodulation
rxI = real(rx) * sqrt(10);
rxQ = imag(rx) * sqrt(10);

Ilevels = [-3 -1 1 3];
Qlevels = [-3 -1 1 3];

Iout = zeros(N,1);
Qout = zeros(N,1);

for k = 1:N
   [dummy, idx] = min(abs(rxI(k) - Ilevels));
   Iout(k) = idx - 1;

   [dummy, idx] = min(abs(rxQ(k) - Qlevels));
   Qout(k) = idx - 1;
end

data_rx = Qout * 4 + Iout;

% Calculate symbol error rate
errors = sum(data ~= data_rx);
SER = errors/N;

fprintf('Symbol Error Rate (SER) = %.4f\n', SER);

% Plot results
figure;

subplot(2,2,1);
scatter(real(tx), imag(tx), 'filled');
title('16-QAM Transmitted Constellation');
xlabel('In-Phase');
ylabel('Quadrature');
grid on;

subplot(2,2,2);
scatter(real(rx), imag(rx), 'filled');
title('Received Constellation with AWGN');
xlabel('In-Phase');
ylabel('Quadrature');
grid on;
```

```
subplot(2,2,[3 4]);
hold on;

sps = 8;
signal = repelem(real(rx), sps);
L = 2*sps;

for k = 1:100
   start = (k-1)*L + 1;
   if start + L - 1 <= length(signal)
      plot((0:L-1)/sps, signal(start:start+L-1));
   end
end

title('Eye Diagram');
xlabel('Time (Symbol Periods)');
ylabel('Amplitude');
grid on;
hold off;
```

OUTPUT